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SM6P07 Digital Media Project

Project

2024-25 Autumn

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One drive Link:

DIGITAL-MEDIA-PROJECT

Github link

<https://github.com/Anupam22067426>

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Abstract

This document presents a comprehensive, detailed, research-based project on the design and development of a full first-person shooter game with 3d environment optimized for the game. The primary aim of the project is to investigate and show the game for the game industry. It also considers the huge market implications of environment design and such as optimization, user engagement and visual storytelling. Key area of focus includes project planning, asse creation, lighting, texturing and real time rendering techniques. In addition to addressing the creative and technical challenges involved. This project aspires to offer important insights and practical knowledge. The goal of this project is to create a game ready environment that supports real time rendering.

Introduction

The upcoming project primarily focuses on exploring on the topic of how meshes are optimized for the games by using tools and 3d software. Game development is increasingly being relied on the efficiency and performance of the processor. The aim of the project is to create a detailed 3d environment while ensuring optimization for real time application and rendering.

This project will utilize Maya and Adobe Substance painter to model and texture the environment to create detailed and visually appealing assets. It will use methods and techniques like reducing the polycount, baking of mesh from high poly to low poly by baking textures and implementing level of detail (LOD) systems to balance visual detail and performance.

The goal of this project is to provide insights into creating efficient, high quality game environments with use of as less polygon as possible. Ultimately, it shows the importance of balancing visually appealing detail with performance friendly environment and optimized for different hardware platforms.

2. Project Overview

This project is based on the research and creation of a game ready environment inspired by modern first-person shooter games. The environment which is designed a immersive setting that gives user a first person experience. The project involves research into the game development process and technical tool

Research question

How is 3d mesh optimized for the game?

Optimization of the 3d mesh is very necessary and standard for the industry. To optimize the mesh, we start by reducing the number of the polygon. Artist still must maintain the shape and geometry of model. We can reduce the number of polygons by deleting the unnecessary geometry like faces and edge also in some cases bevel edges and instead use crease on those edges. We still need to have details on the mesh so instead of using the model with high poly for more detail we first bake the high poly and reduce the number of polygons of high poly and making it low poly and then projecting the texture of high poly mesh into low poly mesh. Artists must keep in mind that for the baking of textures low poly mesh must have proper UV for it to keep the similar textures from the high poly.

Artist also uses the technique like optimizing the topology of the mesh by localising topology, using triangles where deformation is not needed and avoiding the n-gons. In some of the game we have seen that there is prebaked lighting on some of the model to reduce the load on GPU. In some software's like Zbrush there are tools such as decimation which directly reduces the number of polygons without reducing the detail,

in other software's like maya and blender there is option of retopology which helps to streamline the process.

Objectives of Project

To design and develop a realistic and playable environment for game

To apply a storytelling technique in the game to enhance the viewig experience

To focus on the optimized level layout for tactical gameplay and fluid player movement

To maintain a smooth gameplay experience

To experiment with lighting and texturing and sound placement

2 The primary objective of this project is to create a semi realistic 3d environment that is based on the structure of office which in mechanics is like other games like valorant, csgo and call of duty. The environment seeks to enhance the understanding of the game design process and entire pipeline that shows both technical and creative skill.

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Project Scope and deliverables

The scope of this project is to build a 3d game environment that looks realistic and is fun to explore which is based on the environment of an office. It will be a small, urban-style map made for shooting - type game, inspired by popular titles like call of duty and CS:GO and valorant. The environment will include streets, building and objects like barricade and wardrobe along with different kind of table and chairs. The goal is to show skill in modeling, texturing, lighting and scene setup using the Arnold and other game development tools.

The final product will include a fully built 3d scene in maya. Screenshots, video and actual project file will be provided as proof of work. There will also be a report of the entire project explaining how the project was done and all the challenges that were faced in the making of the project. This project will also be used as a portfolio piece that can help in future job applications in the game design and or 3d art industry.

Target Audience

Interest: competitive gamers who enjoy the skill-based game.

Age: There will be some violent element like gun 16 – 35 ages.

Gender: mostly targeted towards males but females can play it too.

Geography: all around the globe because the concept will be familiar and as it is inspired from well-known game.

User experience: Players can feel really connected as the game will require a lot of focus and strategy and teamwork. Player can also feel some challenge as they will be playing against other real human with almost same amount of experience

3. Market Research

Research

The upcoming project primarily focuses on exploring on the topic of how meshes are optimized for the games by using tools and 3d software. Game development is increasingly being relied on the efficiency and performance of the processor. The aim of the project is to create a detailed 3d environment while ensuring optimization for real time application and rendering.

This project will utilize Maya and Adobe Substance painter to model and texture the environment to create detailed and visually appealing assets. It will use methods and techniques like reducing the polycount, baking of mesh from high poly to low poly by baking textures and implementing level of detail (LOD) systems to balance visual detail and performance.

The goal of this project is to provide insights into creating efficient, high quality game environments with use of as less polygon as possible. Ultimately, it shows the importance of balancing visually appealing detail with performance friendly environment and optimized for different hardware platforms.

One of the most important things to remember while making models and props for the game is the proper optimized topology. Game topology usually means that we must get as much detail as possible with as low poly count as possible. Lower the poly count the smoother the game will be. When we lose polycount it becomes difficult to capture all the detail of the model. In that time, we use texturing for the detail capture. It is very important for game to have as low topology as possible as it will take less time to

render, and game will be more efficient.

Game topology also depends on the type of assets that we are dealing with. For example, when it comes to building and other models that won't be animated in the production, we can minimize the polygon and keep clean geometry. Poorly designed model with unnecessary complex topology can result in less optimized game and poor experience for the user.

Some of the software like unreal and Maya support (LOD) systems in which the high-resolution version of the mesh can be replaced by the low-resolution mesh at a distance so it can be modelled in low poly count

For this project I looked at the popular games like CS: GO, call of duty and valornt design and their map. i paid attention to how the levels are built what kind of props they use in their game and how lighting are used in the project like this

Product Review

The game environment is that I am going to model is inspired by the games like call of duty and counter - strike, based on that the environment will be semi- realistic.

Counter – Strike

The environment of Counter-Strike looks semi-realistic due to excellent use of textures and object scale.

Many reusable assets, including barrels and barriers, are found in the map.

The game is optimized, with perfect topology and excellent use of normal maps and baked textures.

All weapons and buildings have clean topology and very low poly count, aiding real-time rendering.

CS:GO stands as an excellent example and case study for any artist.

Pros

Some of the maps of the counter strike have realistic design standard which helps for visually appealing and functional gameplay

It has got interactive environments such as breakable props which makes it very player friendly

It teaches us the performance driven design for smooth and fast paced gameplay across much hardwares.

Since it is online multiplayer, game can be played again and again.

Cons

Many of the texture in the game are repeated which can make it seem repetitive

Just like the texture many of the building are also repeated which can also make the architecture repetitive

There is lack of atmospheric elements like dynamic weather in the game

Since it follows more realism theme the colour palette used for the game is restricted to dull colour to simulate the realife

Figure 1: counter strike

Call of duty

Call of Duty is a benchmark for first-person shooting games with high-detail assets and immersive environments.

The game uses low-poly and optimized poly counts, mainly for weapons and characters.

Unlike Counterstrike, it emphasizes high-end textures, utilizing PBR materials for realistic surfaces.

Call of Duty focuses more on large-scale environment building.

It extensively uses modular buildings to reduce artist workload and maintain clean workflows.

Pros

Call of duty utilizes PBR based rendering and ray tracing in the game so it gives realistic lighting

Game does a very good job in conveying the story with the help of props and assets

Unlike counter strike call of duty has a large variety of props include some unique and some reusable props

Like props there is also a large variety of building in the game which makes it less repetitive

Cons

It has hyper detailed assets and environment, so it requires more time, tools and technical expertise.

There are some high polygon assets despite being a game so it can be a technical challenge during performance maintenance

There are some physics simulations that can be found which can create challenge for maintaining performance

Since it is based around the fast paced gameplay players often overlook complex detail so the artist talent may be ignored

Figure 2: call of duty

Valorant

The valorant game has the balance use of realistic proportions with vibrant and clean designs.

Maps are designed with distinct colour schemes and has clear shapes which the artist focus on making it functional and easy for plyers to understand.

Each map has its own unique theme which offers variety while still fitting into the game world

Environment are designed for competitive play using simple model and detailed textures

Although with great visuals on maps it does not include strong storytelling which can feel limiting to the artist who tend to add narration in their game.

Pros:

Valorant being a first perspective game has a combination of vibrant style and realism. The navigation and the design principles help the user to guide through the game in a easy way.

Maps and characters used in the game are bold and use contrast colours that enhance the clarity

Lighting is designed in such a way that it highlights the overall area and the key area of the game.

The maps have a stye of unique storytelling which adds the narrative details into the environment design

Cons:

The materials used in the design lack details which may seem to be not appealing to the user.

The props used in the game design ae used frequently. This shows how the variety of design has limitation to the game\

4. Project Plan

Project Timeline

p

Digital Media Project

Week Ending

Week 1

Week 2

Week 3

Week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

Week 13

Week 14

Week 15

Week 18

Week 17

Week 19

Week 20

Week 21

Week 22

Week 23

Week 24

p

Concept Development

p

Reference and mood board

p

Style and tools research

p

Basic Blockout

p

Blockout review and refinement

p

Environment modelling

..

Prop modelling

p

UV and optimization

p

Texturing main environment

..

Texturing props and material setup

p

Lighting setup

..

Test renders and adjustment

..

Polish works

..

Revision based feedback

Tools and Technologies Used

Technology of choice

Platform: PC (windows)

Software Used: Autodesk maya: for 3d modelling, rendering, UV mapping)

Substance Painter (Texturing)

Photoshop (2D texture)

Focus of technology in this project was of Autodesk maya because I am already familiar with it and its good for making game assets. I used substance painter for texturing since it makes the model look more real and detailed. After rendering I used premiere pro to convert all the image sequence into video.

Resources Used

Hardware: ASUS ROG G16 (3060 ti) (1TB)

Software: Autodesk maya, Substance Painter, Photoshop

Production skill: Modelling, Texturing, Optimizing, Lighting

Research: Maps from Call of Duty and CS: GO

Collaboration or Manpower

I completed all the project on my own from planning to modelling to texturing to lighting to camera I did all of it. I was responsible for for all of the aspects. Most of the decision, such as design choices, tools and workflow, were made on my own. My supervisor provided some of the guidance early on, mainly for research and layout improvements, but I handled the rest. This project allowed me to apply what I learned in 3d in all of my bachelors program to one single project which pushed my 3d skill further. The experience was hands on and helped me developed a better understanding of the process, from brainstorming phase to the completion of the entire project by myself. The feedback I received helped me refine my approach, but in the end, the success of the project was the result of my effort with the guidance of my supervisor.

Distribution Platform

Since this is a student project and not a full-fledged market ready game it can't be found in any platform, expect it GitHub I have used GitHub to track the progress the entire time. The environment will be presented through a video walkthrough and high-quality screenshots of the entire production process. This video will be shared on the YouTube. YouTube is useful for showing how the environment looks in real time, especially for lighting and mood.

Later I have decided that I will import the project in unreal engine where it can be explored by everyone. This also makes it easier to use the project as portfolio piece. The goal is to make the environment accessible and easy to view also to understand. It is still very important to show the project in a professional manner

5. Production Process

Pre-Production

Initial ideas concept

The initial concept of the project was to design a semi realistic 3d game environment made for the strategic combat which was inspired from the game like counter strike and call of duty. These games offer a very different kind of experience than other game of same genre. The goal was to fuse all these characteristics and make an environment for the game. This direction of concept set the tone for the remainder of the project, making sure all the design choices from layout to the lighting- supported a unified gameplay experience. The setting would incorporate elements such as hallways and rooms including other things such as ward board, tables and barriers and interior room. Ultimately the concept aimed to combine real world believability with engaging gameplay flow, giving players both tactical options and visual interest throughout the map.

Mood board

This mood board shows a clean, modern corporate setting for a tactical FPS game. It includes eight images of office buildings, quiet lobbies, stairwells, and boardrooms. The colors are mostly gray, blue, and neutral tones, giving everything a sleek and serious look. There are no people in the scenes, which makes the spaces feel calm but a little tense like something could happen at any moment. The buildings have a sharp, professional design that feels organized and high-tech. Overall, it sets the mood for action in a place that looks controlled on the outside but could turn dangerous fast. Color palette leaned toward more dark and grey tones to sort of show the ambience of the war. These boards were very important while modeling and texturing as they helped to create consistency in the overall scenes of the project. Mood board played a very crucial role in the overall making of the project

 Figure 3: Moodboard 1

Figure 4: Moodboard 2

Production

Modelling Process

Chair



Figure 5: Modelling process of chair **1**



Figure 6 : Modelling process of chair 2

Figure 7: Modelling process of chair 3

Figure 8: Modelling process of chair 4

Figure 9: Final chair

Sofa



Figure 10: Modelling process of sofa 1



Figure 11: Modelling process of sofa 2

Figure 12: Modelling process of sofa 3

Figure 13: Modelling process of sofa 4

Figure 14: final sofa

Coffee Table



Figure 15: Modelling process of coffee table1

Figure 16: Modelling process of coffee table 2

Figure 17: Modelling process of coffee table 3

Figure 18: final model of coffee table

Mini fridge



Figure 19: Modelling process of minifridge **1**



Figure 20: Modelling process of mini fridge 2

Figure 21: Modelling process of mini fridge 3

Figure 22: final model of mini fridge

Wardrobe



Figure 23: Modelling process of wardrobe **1**



Figure 24: Modelling process of wardrobe 2



Figure 25: Modelling process of handle 1



Figure 26: Modelling process of handle 2

Bar Chair



Figure 27: Modelling process of bar chair **1**



Figure 28: Modelling process of bar chair 2

Figure 29: Modelling process of bar chair 3

Figure 30: Modelling process of bar chair 4

Figure 31: Modelling process of bar chair 5

Figure 32: Modelling process of bar chair 6

Figure 33: Final model of the bar chair

Table



Figure 34: Modelling process of Table 1



Figure 35: Modelling process of Table 2

Figure 36: Modelling process of Table 3

Figure 37: Modelling process of Table 4

Barrel



Figure 38: Modelling process of barrel 1



Figure 39: Modelling process of barrel 2

Figure 40: Modelling process of barrel 3

Figure 41: Modelling process of barrel 4

Figure 42: Modelling process of barrel 5

Figure 43: Final model of the barrel

Book



Figure 44: Modelling process of book 1



Figure 45: Modelling process of book 2

Figure 46: Modelling process of book 3

Figure 47: Final model of the book

UV unwrapping

Uv unwrapping is one of the most important processes of in the 3d production pipeline for the process of UV unwrapping I used maya's Automatic Uv unwrapping

Barrel

Figure 48: Uv of Barrel 1

Figure 49: Uv of Barrel 2

Figure 50: Uv of barrel 3

Bar chair

Figure 51: Uv of Bar chair 1

Figure 52: Uv of Bar chair 2

Coffee Table



Figure 53: Uv of Coffee table 1



Figure 54: Uv of coffee table 2

Books

Figure 55: Uv of Book 1

Figure 56: Uv of Book 2

Mini fridge

Figure 57: Uv of Mini fridge 1

Figure 58: Uv of Mini fridge 2

Office Table

Figure 59: Uv of office table 1

 19

Figure 60: Uv of Office table 2

Shelf

Figure 61: Uv of Shelf 1

Figure 62: Uv of shelf 2

Sofa

Figure 63: Uv of sofa 1

Figure 64: Uv of sofa 2

Wardrobe

Figure 65: Uv of wardrobe 1

Figure 66: Uv of wardrobe 2

Chair

Figure 67: Uv of chair 1

Figure 68: Uv of chair 2

Texturing

Bar chair

Figure 69: Texturing of Bar chair 1

Figure 70: Texturing of Bar chair 2

Figure 71: Texturing of Bar chair 3

Figure 72: Texturing of Bar chair 4

Figure 73: Texturing of Bar chair 5

Coffee Table



Figure 74: Texturing of Coffee table 1



Figure 75: Texturing of Coffee table 2

Figure 76: Texturing of Coffee table 3

Figure 77: Texturing of Coffee table 4

Figure 78: Texturing of Coffee table 5

Mini Fridge

Figure 79: Texturing of Mini fridge 1

Figure 80: Texturing of Mini fridge 2

Figure 81: Texturing of Mini fridge 3



Figure 82: Texturing of Mini fridge 4



Figure 83: Texturing of Mini fridge 5

Office Table

Figure 84: Texturing of Office Table 1

Figure 85: Texturing of Office Table 2

Figure 86: Texturing of Office Table 3

Figure 87: Texturing of Office Table 4

Shelf

Figure 88: Texturing of shelf 1

Figure 89: Texturing of shelf 2

Figure 90: Texturing of shelf 3

Figure 91: Texturing of shelf 4

Sofa

Figure 92: Texturing of Sofa 1

Figure 93: Texturing of Sofa 2

Figure 94: Texturing of Sofa 3

Figure 95: Texturing of Sofa 4

Figure 96: Texturing of Sofa 5

Wardrobe

Figure 97: Texturing of Wardrobe 1

Figure 98: Texturing of Wardrobe 2

Figure 99: Texturing of Wardrobe 3

Chair

Figure 100: Texturing of Chair 1

Figure 101: Texturing of Chair 2

Figure 102: Texturing of chair 3

Post production

Render of the Environment

For the final output of the project, I decided to render the environment in HD with resolution of 1920 x1080 with the using of the Arnold render engine which is a plugin for maya. I chose this resolution because its full HD which means its clear enough to show all the detail which is present in the scene including all the shadows and texture. The face count of the scene is very low as it is a game environment which I think is very important. I tried to keep things low poly as possible, and I relied more on the texturing part and use of maps such as bump and normal maps. The render took quite a time. All of the rendered image and video can be found inside the file.

6. Conclusion

This project helped me improve my 3d skill like modelling texturing and lighting and handling everything from concept to the final render on my own. Early feedback from the supervisor also helped me a lot. Overall, it was a great learning experience

7. Survey results

It was said that the survey form would be taken to see the effective of the final product.

There were 15 questions, name and email address are not revealed in the documentation, but they were asked for any follow up questions. The video link was in the form itself.

 Survey Link:

<https://docs.google.com/forms/d/e/1FAIpQLSdkkAZMsknY6krVJvUKPAjQlwKzDL1V7e41RweXdPJMATvADg/viewform>

Reflection on the Survey

Looking at the survey results, its clear that the project was generally liked. Most people thought the layout and lighting and texture were done well. However, some responses pointed out areas that could be improved.

Some of the review showed that few people did not find out the environment didn't feel fully immersed. It was also interesting to see while some of the people were curious and engaged some people remained neutral about whether they liked the scenes or not.

Overall the feedback was a valuable remainder that even strong work can benefit from further refinement. It gives me a clearer direction for how to improve on the future project.

7. Weekly logs